

Brainstorming - Idea Generation - Prioritization

Phase 1: Brainstorming and Ideation Phase

shopEZ E-commerce Marketplace | MERN Stack | June 2026

1. Problem Statement Identification

Small and medium-sized retail businesses struggle to set up and manage an online storefront efficiently due to complex setups, high subscription fees, and lack of customization. Customers demand a fast, frictionless, and secure online shopping experience with persistent carts, advanced filters, transparent pricing, and trustable payments.

PS-1: The Online Shopper (Convenience and Trust)

- **User Profile:** A busy online consumer seeking smooth retail experience across mobile and desktop.
- **Objective:** Find and purchase items quickly with minimal friction.
- **Obstacles:** Cart items disappearing on session reload, slow page loads, complex checkout pipelines, no real-time inventory checks.
- **Negative Consequences:** Friction triggers frustration, resulting in high cart abandonment rates.

PS-2: The Store Admin / Seller (Operations and Control)

- **User Profile:** Business owner or manager handling catalog and orders.

- **Objective:** Manage product stocks, process orders, review customer feedback, and verify sales growth metrics.
- **Obstacles:** Clunky database updates, difficulty uploading assets, and manual order status tracking.
- **Negative Consequences:** Inventory desyncs and delayed fulfillments.

2. Brainstorming and Idea Listing

The development team collaborated to list potential solutions and architectural requirements:

1. **Custom-built MERN Stack Application:** Build a fully decoupled stack (React, Node, Express, MongoDB) for absolute control over UI, performance, and backend logic.
2. **JWT-based Stateless Authentication:** Implement JSON Web Tokens stored in HTTP-Only cookies to protect against XSS and maintain secure sessions.
3. **Role-Based Access Control (RBAC):** Build distinct user hierarchies (Customer vs. Seller/Admin) into API and UI routers.
4. **Persistent Shopping Cart:** Store cart states in React state (syncd to localized context) and associate order logs in MongoDB.
5. **Interactive Admin Ledger:** Create visual charts (Chart.js) to display monthly sales, revenue progress, and real-time low-stock alarms.
6. **Local Multer Image Storage:** Multer disk storage for product image uploads with static file serving.
7. **Mock Stripe Payment Intents:** Simulated payment endpoint returning client secrets to model industry-standard gateway checkout.

3. Categorization and Grouping of Ideas

Category	Ideas
Core Architecture and Security	Full-stack MERN decoupling; JWT token-based authentication; Middleware enforcing access levels (user, seller, admin)
Customer Experience (CX)	Auto-saving persistent shopping cart; Catalog search with autocomplete suggestions; Multi-step checkout with coupon deductions; Verified purchase review badge and upvoting

Administrative Operations	Centralized dashboard with sales stats; Instant product CRUD operations; Customer status controllers (block/unblock); Active coupon rules compiler
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4. Priority Matrix (MoSCoW Method)

Ideas were prioritized to ensure the core MVP could be delivered before layering secondary and advanced enhancements:

Must Have (High Priority)	Should Have (Medium Priority)	Could Have (Low Priority)	Won't Have (Deferred)
JWT Authentication and RBAC	Admin Dashboard Sales Charts	Related Product Suggestions	Multi-vendor commission splits
Product Catalog Listing	Coupon Code Validation Engine	Wishlist One-Tap Move to Cart	Advanced AI recommendations
Persistent Shopping Cart	Customer Review Upvotes	Shipping Speed Selectors	Multi-currency conversions
Multi-Step Checkout Flow	Low Stock Notification Alerts	Address Book Managers	Native mobile applications
Mock Stripe Payments	Verified Buyer Review Badge	Account Info Editor	Live chat customer support